

Introduction to Machine Learning

Problems: Convolutional Neural Networks

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1. *Tensors*. For each of the following datasets, describe how you would represent them as tensors. Specifically, give the shape of the tensors.
 - (a) A batch of 100 color images, each image is 256×256 .
 - (b) A batch of 40 EEG recordings. Each EEG records has 80 channels of output at a sample rate of 240 Hz for 10 seconds.
 - (c) A batch of 32 videos. Each video has a frame rate of 30 frames per second and is 10 seconds long. The video is color with a resolution of 512×512 .

Solution:

- (a) For a batch of 100 color images, each 256×256 , the tensor shape is $(100, 256, 256, 3)$. The dimensions represent: batch size, height, width, and RGB channels.
- (b) For 40 EEG recordings with 80 channels at 240 Hz for 10 seconds: The number of time samples is $240 \times 10 = 2400$. The tensor shape is $(40, 2400, 80)$. The dimensions represent: batch size, time samples, and channels.
- (c) For 32 videos at 30 fps for 10 seconds with color resolution 512×512 : The number of frames is $30 \times 10 = 300$. The tensor shape is $(32, 300, 512, 512, 3)$. The dimensions represent: batch size, frames (time), height, width, and RGB channels.

2. *2D convolutions*. Let X and W be arrays,

$$X = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 3 & 3 & 0 \\ 0 & 3 & 3 & 3 & 0 \\ 0 & 3 & 2 & 3 & 0 \\ 0 & 3 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}.$$

Let Z be the 2D convolution (without reversal):

$$Z[i, j] = \sum_{k_1, k_2} W[k_1, k_2] X[i + k_1, j + k_2]. \quad (1)$$

Assume that the arrays are indexed starting at $(0, 0)$.

- (a) What are the limits of the summations over k_1 and k_2 in (1)?

- (b) What is the size of the output $Z[i, j]$ if the convolution is computed only on the *valid* pixels (i.e. the pixel locations (i, j) where the summation in (1) does not exceed the boundaries of W or X).
- (c) What is the largest positive value of $Z[i, j]$ and state one pixel location (i, j) where that value occurs.
- (d) What is the largest negative value of $Z[i, j]$ and state one pixel location (i, j) where that value occurs.
- (e) Find one pixel location where $Z[i, j] = 0$.

Solution:

- (a) The kernel W has shape $(2, 2)$, so $k_1 \in \{0, 1\}$ and $k_2 \in \{0, 1\}$.
- (b) For valid convolution, we need $i + k_1 < 6$ and $j + k_2 < 5$ for all $k_1 \in \{0, 1\}$, $k_2 \in \{0, 1\}$. This means $i \leq 4$ and $j \leq 3$. The output Z has size $(5, 4)$.
- (c) Computing $Z[i, j]$ for valid positions:

$$\begin{aligned}
Z[0, 0] &= 1 \cdot 0 + (-1) \cdot 0 + 1 \cdot 0 + (-1) \cdot 3 = -3 \\
Z[0, 1] &= 1 \cdot 0 + (-1) \cdot 0 + 1 \cdot 3 + (-1) \cdot 3 = 0 \\
Z[0, 2] &= 1 \cdot 0 + (-1) \cdot 0 + 1 \cdot 3 + (-1) \cdot 3 = 0 \\
Z[0, 3] &= 1 \cdot 0 + (-1) \cdot 0 + 1 \cdot 3 + (-1) \cdot 0 = 3 \\
Z[1, 0] &= 1 \cdot 0 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 3 = -3 \\
Z[1, 1] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 3 = 0 \\
Z[1, 2] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 2 = 1 \\
Z[1, 3] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 2 + (-1) \cdot 3 = -1 \\
Z[2, 0] &= 1 \cdot 0 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 3 = -3 \\
Z[2, 1] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 3 = 0 \\
Z[2, 2] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 2 = 1 \\
Z[2, 3] &= 1 \cdot 3 + (-1) \cdot 3 + 1 \cdot 2 + (-1) \cdot 3 = -1 \\
Z[3, 0] &= 1 \cdot 0 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 2 = -2 \\
Z[3, 1] &= 1 \cdot 3 + (-1) \cdot 2 + 1 \cdot 3 + (-1) \cdot 2 = 2 \\
Z[3, 2] &= 1 \cdot 2 + (-1) \cdot 3 + 1 \cdot 2 + (-1) \cdot 3 = -2 \\
Z[3, 3] &= 1 \cdot 3 + (-1) \cdot 0 + 1 \cdot 3 + (-1) \cdot 0 = 6 \\
Z[4, 0] &= 1 \cdot 0 + (-1) \cdot 3 + 1 \cdot 3 + (-1) \cdot 2 = -2 \\
Z[4, 1] &= 1 \cdot 3 + (-1) \cdot 2 + 1 \cdot 3 + (-1) \cdot 2 = 2 \\
Z[4, 2] &= 1 \cdot 2 + (-1) \cdot 3 + 1 \cdot 2 + (-1) \cdot 3 = -2 \\
Z[4, 3] &= 1 \cdot 3 + (-1) \cdot 0 + 1 \cdot 0 + (-1) \cdot 0 = 3
\end{aligned}$$

The largest positive value is $Z[3, 3] = 6$ (and also $Z[4, 3] = 6$).

- (d) The largest negative value is $Z[0, 0] = -3$ (and also $Z[1, 0] = -3$ and $Z[2, 0] = -3$).
- (e) One location where $Z[i, j] = 0$ is $(0, 1)$, $(0, 2)$, or $(1, 1)$.

3. *Complexity and number of parameters.* Suppose that a convolutional layer of a neural network has an input tensor $X[i, j, k]$ and computes an output via a convolution and ReLU activation,

$$Z[i, j, m] = \sum_{k_1} \sum_{k_2} \sum_n W[k_1, k_2, n, m] X[i + k_1, j + k_2, n] + b[m],$$

$$U[i, j, m] = \max\{0, Z[i, j, m]\}.$$

for some weight kernel $W[k_1, k_2, n, m]$ and bias $b[m]$. Suppose that X has shape $(48, 64, 10)$ and W has shape $(3, 3, 10, 20)$. Assume the convolution is computed on the *valid* pixels.

- (a) What are the shapes of Z and U ?
- (b) What are the number of input channels and output channels?
- (c) How many multiplications must be performed to compute the convolution in that layer?
- (d) If W and b are to be learned, what are the total number of trainable parameters in the layer?

Solution:

- (a) The input X has shape $(48, 64, 10)$ representing (height, width, input channels). The kernel W has shape $(3, 3, 10, 20)$ representing (kernel height, kernel width, input channels, output channels).

For valid convolution with a 3×3 kernel:

$$\text{Output height} = 48 - 3 + 1 = 46$$

$$\text{Output width} = 64 - 3 + 1 = 62$$

$$\text{Output channels} = 20$$

Therefore, both Z and U have shape $(46, 62, 20)$.

- (b) The number of input channels is 10, and the number of output channels is 20.
- (c) For each output position (i, j) and each output channel m , we compute:

$$Z[i, j, m] = \sum_{k_1=0}^2 \sum_{k_2=0}^2 \sum_{n=0}^9 W[k_1, k_2, n, m] X[i + k_1, j + k_2, n] + b[m]$$

This requires $3 \times 3 \times 10 = 90$ multiplications per output element.

The number of output elements is $46 \times 62 \times 20 = 57,040$.

Total multiplications: $57,040 \times 90 = 5,133,600$.

(d) The number of trainable parameters:

$$\text{Parameters in } W : 3 \times 3 \times 10 \times 20 = 1,800$$

$$\text{Parameters in } b : 20$$

$$\text{Total} : 1,820$$

4. *Back-propagation.* Suppose that a convolutional layer in some neural network is described as a linear convolution followed by a sigmoid activation,

$$Z[i, j_1, j_2, m] = \sum_{k_1} \sum_{k_2} \sum_n W[k_1, k_2, n, m] X[i, j_1 + k_1, j_2 + k_2, n] + b[m],$$

$$U[i, j_1, j_2, m] = 1/(1 + \exp(-Z[i, j_1, j_2, m])).$$

where $X[i, j_1, j_2, n]$ is the input of the layer and $U[i, j_1, j_2, m]$ is the output. Suppose that during back-propagation, we have computed the gradient $\partial J/\partial U$ for some loss function J . That is, we have computed the components $\partial J/\partial U[i, j_1, j_2, m]$. Show how to compute the following:

- (a) The gradient components $\partial J/\partial Z[i, j_1, j_2, m]$.
- (b) The gradient components $\partial J/\partial W[k_1, k_2, n, m]$.
- (c) The gradient components $\partial J/\partial X[i, j_1, j_2, n]$.

Solution:

- (a) Using the chain rule, we have:

$$\frac{\partial J}{\partial Z[i, j_1, j_2, m]} = \frac{\partial J}{\partial U[i, j_1, j_2, m]} \cdot \frac{\partial U[i, j_1, j_2, m]}{\partial Z[i, j_1, j_2, m]}$$

Since $U[i, j_1, j_2, m] = \sigma(Z[i, j_1, j_2, m]) = 1/(1 + \exp(-Z[i, j_1, j_2, m]))$, the derivative is:

$$\frac{\partial U}{\partial Z} = U(1 - U) = \sigma(Z)(1 - \sigma(Z))$$

Therefore:

$$\frac{\partial J}{\partial Z[i, j_1, j_2, m]} = \frac{\partial J}{\partial U[i, j_1, j_2, m]} \cdot U[i, j_1, j_2, m](1 - U[i, j_1, j_2, m])$$

- (b) To compute $\partial J/\partial W[k_1, k_2, n, m]$, we use the chain rule through Z :

$$\frac{\partial J}{\partial W[k_1, k_2, n, m]} = \sum_i \sum_{j_1} \sum_{j_2} \frac{\partial J}{\partial Z[i, j_1, j_2, m]} \cdot \frac{\partial Z[i, j_1, j_2, m]}{\partial W[k_1, k_2, n, m]}$$

From the convolution equation:

$$Z[i, j_1, j_2, m] = \sum_{k'_1} \sum_{k'_2} \sum_{n'} W[k'_1, k'_2, n', m] X[i, j_1 + k'_1, j_2 + k'_2, n'] + b[m]$$

We have:

$$\frac{\partial Z[i, j_1, j_2, m]}{\partial W[k_1, k_2, n, m]} = X[i, j_1 + k_1, j_2 + k_2, n]$$

Therefore:

$$\frac{\partial J}{\partial W[k_1, k_2, n, m]} = \sum_i \sum_{j_1} \sum_{j_2} \frac{\partial J}{\partial Z[i, j_1, j_2, m]} \cdot X[i, j_1 + k_1, j_2 + k_2, n]$$

- (c) To compute $\partial J / \partial X[i, j_1, j_2, n]$, we need to sum over all positions where $X[i, j_1, j_2, n]$ appears in the convolution. For a given input position (i, j_1, j_2, n) , it contributes to $Z[i, j'_1, j'_2, m]$ when $j'_1 + k_1 = j_1$ and $j'_2 + k_2 = j_2$, i.e., when $j'_1 = j_1 - k_1$ and $j'_2 = j_2 - k_2$.

$$\frac{\partial J}{\partial X[i, j_1, j_2, n]} = \sum_{k_1} \sum_{k_2} \sum_m \frac{\partial J}{\partial Z[i, j_1 - k_1, j_2 - k_2, m]} \cdot \frac{\partial Z[i, j_1 - k_1, j_2 - k_2, m]}{\partial X[i, j_1, j_2, n]}$$

Since $\partial Z[i, j_1 - k_1, j_2 - k_2, m] / \partial X[i, j_1, j_2, n] = W[k_1, k_2, n, m]$:

$$\frac{\partial J}{\partial X[i, j_1, j_2, n]} = \sum_{k_1} \sum_{k_2} \sum_m \frac{\partial J}{\partial Z[i, j_1 - k_1, j_2 - k_2, m]} \cdot W[k_1, k_2, n, m]$$

This is essentially a convolution of the gradient $\partial J / \partial Z$ with the flipped kernel W .

5. *Sub-sampling and pooling.* In CNNs, convolution operations are often followed by a data reduction step, typically either via *sub-sampling* or *max pooling*. The methods can be described as follows: Let $x[j]$, $j = 0, 1, \dots, N - 1$ be a 1D input (say in one channel in one sample). The outputs $y[k]$ for sub-sampling and max-pooling are given by:

- *Sub-sampling* with *stride* s selects every s -th sample:

$$y[k] = x[sk], \quad k = 0, 1, \dots, \left\lfloor \frac{N-1}{s} \right\rfloor.$$

- *Max pooling* with *pool size* p and *stride* s computes,

$$y[k] = \max_{j=0,1,\dots,p-1} x[sk + j], \quad k = 0, 1, \dots, \left\lfloor \frac{N-1}{s} \right\rfloor.$$

- (a) Let $\mathbf{x}$ be the vector,

$$\mathbf{x} = [1, 2, 3, 2, 0, 10, 1, 0].$$

Find the output $\mathbf{y}$ when sub-sampling with stride $s = 2$.

- (b) For the same vector $\mathbf{x}$ as in part (a), find the output of max pooling with stride $s = 2$ and pool size $p = 2$.
- (c) Let $X[i, j, n]$ be a tensor of shape (B, N, C) where B is the batch size, N is the number of samples per channel and C is the number of channels. Write equations for sampling and max pooling of X if the operations are to be performed on each channel and sample. What are the output shapes?

Solution:

- (a) For $\mathbf{x} = [1, 2, 3, 2, 0, 10, 1, 0]$ with sub-sampling stride $s = 2$:

$$y[0] = x[0] = 1$$

$$y[1] = x[2] = 3$$

$$y[2] = x[4] = 0$$

$$y[3] = x[6] = 1$$

Therefore, $\mathbf{y} = [1, 3, 0, 1]$.

- (b) For the same vector $\mathbf{x} = [1, 2, 3, 2, 0, 10, 1, 0]$ with max pooling stride $s = 2$ and pool size $p = 2$:

$$y[0] = \max(x[0], x[1]) = \max(1, 2) = 2$$

$$y[1] = \max(x[2], x[3]) = \max(3, 2) = 3$$

$$y[2] = \max(x[4], x[5]) = \max(0, 10) = 10$$

$$y[3] = \max(x[6], x[7]) = \max(1, 0) = 1$$

Therefore, $\mathbf{y} = [2, 3, 10, 1]$.

- (c) For a tensor $X[i, j, n]$ of shape (B, N, C) :

Sub-sampling with stride s :

$$Y[i, k, n] = X[i, sk, n], \quad k = 0, 1, \dots, \left\lfloor \frac{N-1}{s} \right\rfloor$$

The output shape is $(B, \left\lfloor \frac{N-1}{s} \right\rfloor + 1, C)$.

Max pooling with stride s and pool size p :

$$Y[i, k, n] = \max_{j=0,1,\dots,p-1} X[i, sk + j, n], \quad k = 0, 1, \dots, \left\lfloor \frac{N-1}{s} \right\rfloor$$

The output shape is $(B, \left\lfloor \frac{N-1}{s} \right\rfloor + 1, C)$.

In both cases, the operations are applied independently to each batch element i and each channel n .